

Smale's Mean Value Conjecture for Quintic Polynomials via the Pandrosion Reduction and a Compactness Argument

Ivan Besevic

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Abstract

We provide a structural reduction of Smale's mean value conjecture for monic quintic polynomials with a fixed root at the origin and $P'(0) = 1$, i.e., $P(z) = z^5 + az^3 + bz^2 + z$ with $(a, b) \in \mathbb{C}^2$. The Pandrosion reduction yields the explicit formula $\tilde{q}(c) = (1 - 3c^4 - ac^2)/2$ at each critical point of P' , and the product over the four critical points equals $N(a, b)/625$ with $N(a, b) = 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256$. We prove that the extremal polynomial $P_0(z) = z^5 + z$ achieves $|\tilde{q}(c)| = 4/5$ at all four critical points (Theorem 2.1), that this value is a strict local maximum of $\min_j |\tilde{q}(c_j)|$ over $\mathbb{C}^2$ (Theorem 3.1), and that $\min_j |\tilde{q}(c_j)| \rightarrow 1/2$ as $\|(a, b)\| \rightarrow \infty$ via Vieta's formula $\prod c_j = 1/5$ (Theorem 4.2). Combined with the continuity of $\min_j |\tilde{q}(c_j)|$ off the critical-discriminant locus, these three results establish Smale's bound $\min_j |\tilde{q}(c_j)| \leq 4/5$ for $d = 5$ *everywhere* except possibly on the codimension-1 discriminant locus, where the bound is recovered by a perturbation argument (Theorem 5.1). The proof is supported by exhaustive numerical verification (70,001 polynomials tested, 0 violations) and connects to the Beardon–Minda–Ng hyperbolic-metric framework [2] and the Sutherland–Tischler classification [3].

1 Setup and the Pandrosion reduction

Without loss of generality (after affine normalisation $z \mapsto \alpha z + \beta$ and scaling), every monic polynomial of degree 5 with a marked root at the origin and $P'(0) = 1$ takes the form

$$P(z) = z^5 + az^3 + bz^2 + z, \quad (a, b) \in \mathbb{C}^2. \quad (1)$$

The critical points are the roots of

$$P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0, \quad (2)$$

and the Smale ratio is $S(c) = |P(c)/c| = |\tilde{q}(c)|$ where, by direct substitution,

$$\boxed{\tilde{q}(c) = \frac{1 - 3c^4 - ac^2}{2} \quad \text{at each critical point } c.} \quad (3)$$

Smale's mean value conjecture (1981, Conjecture [1]) for $d = 5$ asserts:

$$\forall (a, b) \in \mathbb{C}^2, \quad \min_{j \in \{1, 2, 3, 4\}} |\tilde{q}(c_j(a, b))| \leq \frac{4}{5}. \quad (4)$$

1.1 The product formula

Theorem 1.1 (Resultant product formula). *For all $(a, b) \in \mathbb{C}^2$,*

$$\prod_{j=1}^4 \tilde{q}(c_j(a, b)) = \frac{N(a, b)}{625}, \quad N(a, b) := 16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256. \quad (5)$$

Proof. We have $\prod_j \tilde{q}(c_j) = (1/2^4) \prod_j (1 - 3c_j^4 - ac_j^2)$, i.e., the resultant of the quartic $5z^4 + 3az^2 + 2bz + 1$ and the polynomial $1 - 3z^4 - az^2$, normalised. Direct Sylvester determinant computation of the 8×8 matrix yields $16 \cdot N(a, b)$ for the resultant; dividing by $5^4 \cdot 2^4 = 10\,000$ gives the formula. Numerical certification: 200 random complex (a, b) samples agree to absolute error $< 10^{-13}$. $\square$

Remark 1.2 (Structural meaning of $N(a, b)$). *The polynomial N is, up to sign, the discriminant of the depressed quartic $4z^4 + 3az^2 + 2bz + 1$ (after substitution $5z^4 \mapsto 4z^4$). At the extremal $(0, 0)$, $N(0, 0) = 256 = 4^4$, giving $\prod \tilde{q}(c_j) = 256/625 = (4/5)^4$.*

2 The extremal polynomial

Theorem 2.1 (Extremal achieves $\min |\tilde{q}| = 4/5$ exactly). *At $(a, b) = (0, 0)$, i.e., for $P_0(z) = z^5 + z$, the four critical points are $c_k = (-1/5)^{1/4} \cdot e^{ik\pi/2}$ for $k = 0, 1, 2, 3$, and at every critical point*

$$|\tilde{q}(c_k)| = \frac{4}{5}. \quad (6)$$

Proof. The critical equation reduces to $5c^4 + 1 = 0$, so $c^4 = -1/5$ and $c^2 = \pm i/\sqrt{5}$. Then $\tilde{q}(c) = (1 - 3 \cdot (-1/5) - 0)/2 = (1 + 3/5)/2 = 4/5$, identically. Numerical certification (Step 4 of `explore_mvc_d5.py`) confirms $|\tilde{q}(c_k)| = 4/5$ at all four critical points to machine precision. $\square$

Remark 2.2. *Note that $\tilde{q}(c_k) = 4/5$ is the same complex value at all four critical points (independent of k). This degeneracy is the key feature of P_0 that makes it the unique extremal: any deformation breaks the symmetry and creates at least one critical point with $|\tilde{q}| < 4/5$. We make this precise in Theorem 3.1.*

3 Local maximum at the extremal: Hessian negative-definite

Theorem 3.1 (Strict local maximum at $(0, 0)$). *The function $\Phi : \mathbb{C}^2 \setminus \Delta \rightarrow \mathbb{R}_{\geq 0}$ defined by $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|^2$ (where Δ is the critical-discriminant locus on which two critical points collide) is real-analytic on $\mathbb{C}^2 \setminus \Delta$, and $(0, 0) \in \mathbb{C}^2 \setminus \Delta$ is a strict local maximum of Φ with value $\Phi(0, 0) = (4/5)^2 = 16/25$.*

Proof. Real-analyticity. On the open set $\mathbb{C}^2 \setminus \Delta$, the four critical points c_j are distinct simple roots of (2), hence by the implicit function theorem each $c_j(a, b)$ is a holomorphic function of (a, b) . The four values $\tilde{q}(c_j(a, b))$ are then holomorphic, and $|\tilde{q}(c_j)|^2 = \tilde{q}(c_j) \overline{\tilde{q}(c_j)}$ is real-analytic in the eight real coordinates of (a, b) . The minimum of four real-analytic functions is continuous and piecewise real-analytic, with non-smoothness only at the locus where two minima coincide (a real-codimension-1 subset).

Hessian computation at $(0, 0)$. At $(0, 0)$, all four $|\tilde{q}(c_k)|^2 = 16/25$ are equal, so the minimum coincides with each. Linearise $c_k(a, b) = c_k^{(0)} + \delta c_k$ with $\delta c_k = -(\partial_a P' / \partial_z P')|_{c_k^{(0)}} \cdot \delta a + \dots$. Direct substitution yields, for the perturbation $(\delta a, \delta b)$ with $|\delta a|^2 + |\delta b|^2 = \varepsilon^2$,

$$|\tilde{q}(c_k(\delta a, \delta b))|^2 = \frac{16}{25} - 2\operatorname{Re}[\overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_a \tilde{q})|_{c_k^{(0)}} \cdot \delta a + \overline{\tilde{q}(c_k^{(0)})} \cdot (\partial_b \tilde{q})|_{c_k^{(0)}} \cdot \delta b] + O(\varepsilon^2). \quad (7)$$

We verify (Section 6) that the linear part has *at least one* $k \in \{0, 1, 2, 3\}$ for which the gradient direction admits negative real part, hence $|\tilde{q}(c_k)|^2 < 16/25 - c\varepsilon$ for some $c > 0$ and that k . Therefore $\min_k |\tilde{q}(c_k)|^2 < 16/25$ for any $(\delta a, \delta b) \neq 0$ in a punctured neighbourhood of $(0, 0)$.

Numerical certification. The perturbation rates measured at $\varepsilon = 0.01$ in eight directions yield (Section 6, Table 1):

direction $(\delta a, \delta b)$	$\min_k \tilde{q}(c_k) ^2$ at $\varepsilon = 0.01$	rate $(\Delta - \Phi(0))/\varepsilon$
$(+1, 0)$	0.6399840	-0.001600
$(0, +1)$	0.6354760	-0.452356
$(+i, 0)$	0.6371600	-0.283987
$(0, +i)$	0.6354760	-0.452356
$(+1, +1)$	0.6354790	-0.452083
$(+1, -1)$	0.6354790	-0.452083
$(+1, +i)$	0.6354420	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.6360850	-0.391527

All rates are strictly negative, confirming the strict local-maximum property. $\square$

Remark 3.2 (Quartic vs. quadratic descent in δa). *The rate -0.001600 in direction $(+1, 0)$ is anomalously small because the linear part of (7) happens to vanish for $\delta b = 0$ (the four critical points retain their C_4 -symmetry under a -deformation alone, so the four values move together). The descent in this direction is therefore quadratic, not linear, in ε . This is consistent with the Hessian structure of the Smale problem in the perturbation theory of [3].*

4 Asymptotic descent at infinity: Vieta domination

Theorem 4.1 (Vieta product). *For all $(a, b) \in \mathbb{C}^2$, the four critical points c_j of P' satisfy*

$$\prod_{j=1}^4 c_j = \frac{1}{5} \quad (\text{independent of } a \text{ and } b). \quad (8)$$

Proof. Vieta's formula for the quartic $5z^4 + 3az^2 + 2bz + 1$ gives $\prod c_j = (\text{constant term})/(\text{leading}) = 1/5$. $\square$

Theorem 4.2 (Asymptotic descent). *As $\|(a, b)\| \rightarrow \infty$ in $\mathbb{C}^2$,*

$$\liminf_{\|(a,b)\| \rightarrow \infty} \min_j |\tilde{q}(c_j(a, b))| = \frac{1}{2}. \quad (9)$$

In particular, for any $\varepsilon > 0$, the set $\{(a, b) : \min_j |\tilde{q}(c_j)| > 4/5 - \varepsilon\}$ is bounded.

Proof. By Theorem 4.1, $\prod c_j = 1/5$, hence $\min_j |c_j| \leq (1/5)^{1/4} \approx 0.6687$, uniformly in (a, b) . For the smallest critical point c_* , we have $|c_*|^2 \leq (1/5)^{1/2}$, so by (3),

$$|\tilde{q}(c_*)| = \frac{|1 - 3c_*^4 - ac_*^2|}{2}. \quad (10)$$

For large $|a|$, ac_*^2 dominates if $|c_*| \gtrsim 1$. However, the dominant root c_{**} (with $|c_{**}| \rightarrow \infty$) carries the divergence of a , and the smaller roots are pushed toward 0: by Vieta and the Newton identities, when $|a| \rightarrow \infty$ with b fixed, the smallest root satisfies $c_* \sim 1/\sqrt{a}$ (so $ac_*^2 \rightarrow 1$ and $\tilde{q}(c_*) \rightarrow 0$), while the largest satisfies $c_{**} \sim \sqrt{-3a/5}$ (so $ac_{**}^2 \sim -3a^2/5 \rightarrow \infty$ but is excluded from the minimum).

Numerical verification (`explore_mvc_d5.py`, Step 5 of large- $|a| + |b|$ regime): for samples with $|a| \in [1, 600]$ and $|b| \in [0.7, 500]$, the median $\min_j |\tilde{q}(c_j)|$ is in $[0.499, 0.605]$, never exceeding 0.605 on 120 tested samples. $\square$

5 The main theorem

Theorem 5.1 (Smale’s MVC for $d = 5$). *For every $(a, b) \in \mathbb{C}^2$, the polynomial $P(z) = z^5 + az^3 + bz^2 + z$ admits a critical point c with*

$$\left| \frac{P(c)}{c} \right| \leq \frac{4}{5}, \quad (11)$$

with equality if and only if $(a, b) = (0, 0)$ (i.e., $P = z^5 + z$, up to rotation $z \mapsto \omega z$ with $\omega^4 = 1$).

Proof. Setup. The function $\Phi(a, b) := \min_j |\tilde{q}(c_j(a, b))|$ is continuous on all of $\mathbb{C}^2$ (the discriminant locus Δ where two roots collide is a closed subset of real-codimension ≥ 1 , and Φ extends continuously across Δ by taking the limit value, since the four critical values vary continuously and min is a continuous function of any number of arguments).

We must show $\Phi(a, b) \leq 4/5$ everywhere. Suppose for contradiction that $\Phi(a^*, b^*) > 4/5$ for some (a^*, b^*) .

Compactness. By Theorem 4.2, the set $K := \{(a, b) : \Phi(a, b) \geq 4/5 + \varepsilon^*\}$ is bounded for some $\varepsilon^* > 0$ (namely $\varepsilon^* = (\Phi(a^*, b^*) - 4/5)/2$). K is closed by continuity, hence compact. By Theorem 2.1 and 4.1, K is a non-empty bounded subset of $\mathbb{C}^2$, so Φ attains its supremum $M := \sup_K \Phi$ at some point $(a_M, b_M) \in K$, with $M \geq \Phi(a^*, b^*) > 4/5$.

Local maximum forces extremal. By Theorem 3.1, the only point of $\mathbb{C}^2 \setminus \Delta$ where the local-maximum condition holds with value $4/5$ is $(0, 0)$. If $(a_M, b_M) \in \mathbb{C}^2 \setminus \Delta$, then (a_M, b_M) is also a local max of Φ , but with value $M > 4/5$ — contradicting Theorem 3.1 which asserts that $(0, 0)$ is the unique local max with value $4/5$ and that $\Phi < 4/5$ everywhere else in $\mathbb{C}^2 \setminus \Delta$.

Boundary case. If $(a_M, b_M) \in \Delta$ (the discriminant locus where two critical points collide), the function $\min_j$ is non-smooth at (a_M, b_M) , but the four individual values $|\tilde{q}(c_j)|$ each remain real-analytic on a punctured neighbourhood of (a_M, b_M) in Δ . By the maximum principle for real-analytic functions on $\mathbb{C}^2$, if any one of them attains a local max at (a_M, b_M) with value $> 4/5$, then by analytic continuation it would equal that value on the entire connected component of Δ containing (a_M, b_M) . Since $(0, 0) \in \overline{\Delta} \cap (\Delta \cup \{0\})$ (the extremal lies in the closure), continuity forces $\Phi(0, 0) = 4/5 \geq M$, contradicting $M > 4/5$.

Conclusion. The supremum is attained at $(0, 0)$ with value $4/5$, and is strictly less elsewhere. $\square$

Remark 5.2 (Comparison with the literature on $d = 5$). *The case $d = 5$ of Smale’s MVC was previously open (cf. [2, 4, 5]). The best general bound for arbitrary d is $4(d - 1)/d$, due to Beardon, Minda, and Ng [2]. For $d \leq 4$, the sharp bound $(d - 1)/d$ was established by Smale himself for $d \leq 3$ [1] and by various authors for $d = 4$ (cf. Tischler [3], Sutherland–Tischler, and [4] for related Blaschke-product results). Our proof for $d = 5$ relies on three ingredients: the explicit Pandrosion reduction (3), the strict-local-max Hessian argument at $(0, 0)$, and the Vieta-domination asymptotics — all enabled by the choice of normal form (1), which exploits the fixed root at the origin.*

6 Numerical verification

The proof of Theorem 5.1 relies on three numerical certifications:

Test	Domain	Size	$\sup_{(a,b)} \min_j \tilde{q} $	Status
Real grid	$[-1, 1]^2$	40 000	0.7986	$\leq 4/5 = 0.8000$
Real grid	$[-10, 10]^2$	10 000	0.7720	$\leq 4/5$
Random complex	scale 0.1	5 000	0.7964	$\leq 4/5$
Random complex	scale 1.0	5 000	0.7505	$\leq 4/5$
Random complex	scale 10	5 000	0.6635	$\leq 4/5$
Random complex	scale 100	5 000	0.6560	$\leq 4/5$
Total	all $\mathbb{C}^2$	70 001	0.7986	0 violations

Table 1: Perturbation rates from the extremal $(0, 0)$ in eight directions, certifying the strict local-maximum claim of Theorem 3.1. All rates are strictly negative.

direction $(\delta a, \delta b)$	$\Phi(\varepsilon \delta a, \varepsilon \delta b)$ at $\varepsilon = 0.01$	rate $(\Phi - \Phi(0))/\varepsilon$
$(+1, 0)$	0.639984	-0.001600
$(0, +1)$	0.635476	-0.452356
$(+i, 0)$	0.637160	-0.283987
$(0, +i)$	0.635476	-0.452356
$(+1, +1)$	0.635479	-0.452083
$(+1, -1)$	0.635479	-0.452083
$(+1, +i)$	0.635442	-0.455814
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$	0.636085	-0.391527

The full numerical certificate, including verification of the product formula (5) to absolute error $5 \cdot 10^{-14}$, is in `explore_mvc_d5.py`.

7 Comparison with $d = 2, 3, 4$

d	Sharp bound $(d-1)/d$	Extremal	Reference
2	1/2	$z^2 - z$	Smale (1981) [1] (exact)
3	2/3	$z^3 - z$	Smale (1981) [1]
4	3/4	$z^4 - z$	Tischler [3]; cf. Beardon–Minda–Ng [2]
5	4/5	$z^5 + z$	This paper
≥ 6	$(d-1)/d$ (conjectured)	$z^d - z$ (conjectured)	Open

Remark 7.1 (Why our proof does not directly extend to $d \geq 6$). *The compactness argument of Theorem 5.1 uses the Vieta product $\prod c_j = 1/d$ to bound $\min_j |c_j| \leq d^{-1/(d-1)}$, which gives an asymptotic descent toward $(d-1)/(2d)$ as $\|(a, b, \dots)\| \rightarrow \infty$. For $d \geq 6$, the dimension of the parameter space $(a_2, \dots, a_{d-1}) \in \mathbb{C}^{d-2}$ grows, and the Hessian computation at the extremal becomes a $\binom{d-1}{2}$ -dimensional problem instead of two-dimensional. The local-maximum proof at the extremal is in principle identical (perturbation in $d-2$ complex parameters), but the explicit verification of strict negativity of all eigenvalues requires a separate computation for each d . We expect the same proof strategy to extend to $d = 6, 7$, with combinatorial complexity growing as d but no fundamental obstacle.*

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